

MIE 313 Course Project - Group 6

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Figure 1 - Final Wrench Design

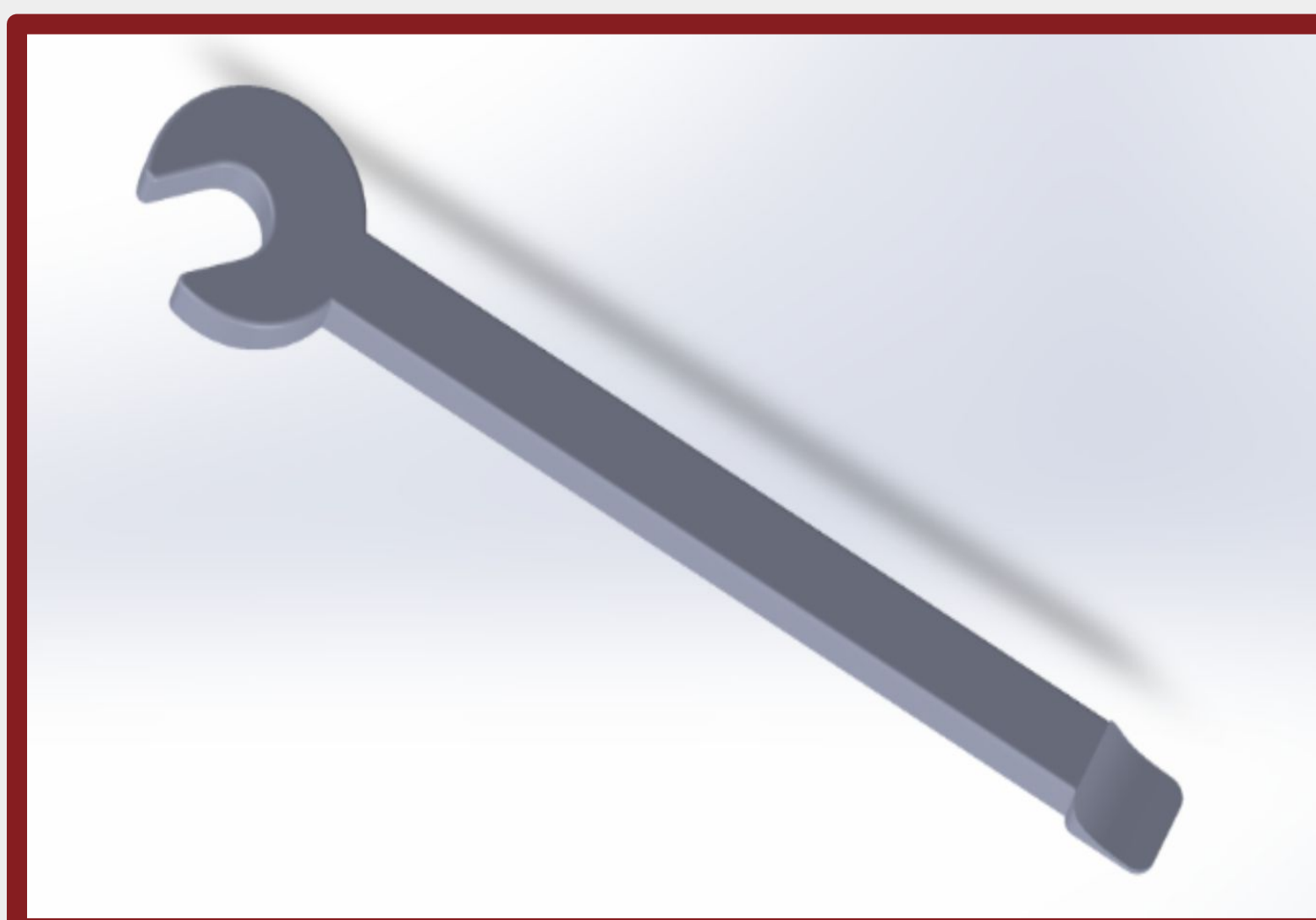


As pictured above, the final design utilizes a sleek I-Beam handle with a closed wrench head. It features a secondary bike tire removal tool. Relevant anticipated dimensions include a max length of 6.55", a max width of 1", and a max thickness of 0.21". Anticipated weight is 31.9g.

Initial Designs:

Pictured right is one initial concept for a wrench design. Though this wrench looks typical and meets the dimensional requirements, it is overweight. In addition, the handle is too long and narrow. The open head causes decreased strength. Finally, the round external profile is impossible to machine.

Figure 2 - Concept Wrench



Project Introduction:

Problem Statement:

Design a lightweight biking wrench for a 1/4-20 bolt that will support 250 in-lb torque, and 50N applied force in a three point bend test without plastic deformation. Cost is but a secondary consideration.

Material Requirement:

Aluminum 6061 material constraint. Inferred relevant properties include a yield strength of 270 Mpa, A Young's modulus of roughly 69 Gpa, and a Poisson's ratio of 0.33.

Size Requirement:

The wrench dimensions must not exceed the size of the aluminum bar stock provided. The dimensions of the bar stock are 1.25" x 0.25" x 8".

Weight Requirement:

The target weight requirement is 1.15 oz (32.6g).

By Hand Analysis Overview:

Figure 3 - Torque Test By Hand Analysis

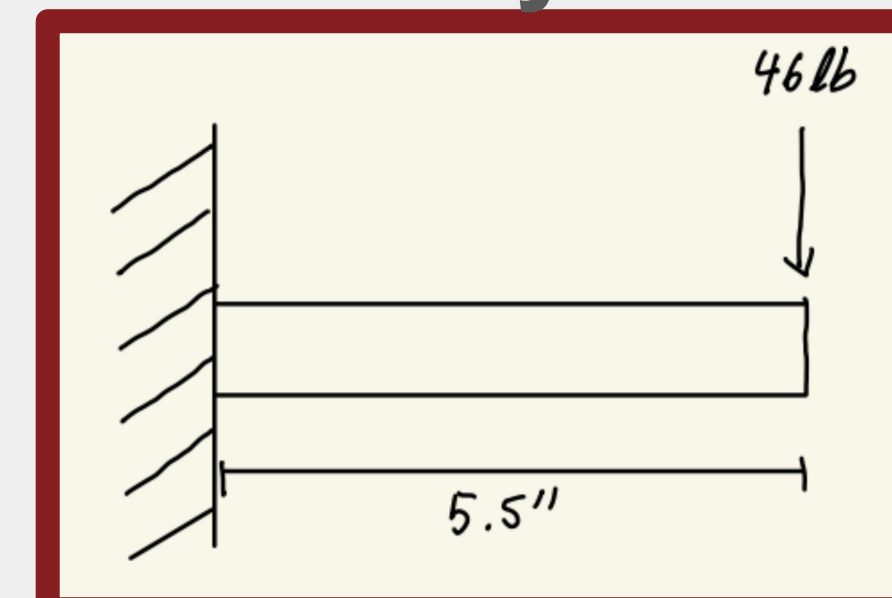
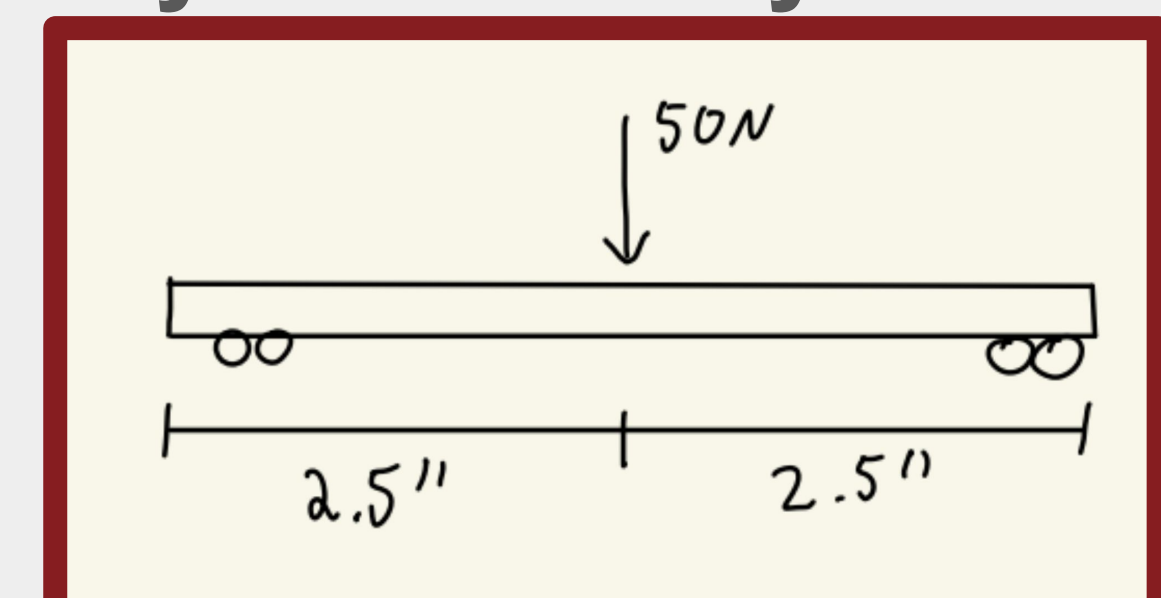


Figure 4 - Bending Test By Hand Analysis



By hand analysis was conducted using Euler-Bernoulli beam theory for the torque test, the combined test, and the bending test. A cross section with a height of .5" and a width of .25" was used. For the torque test, this yielded a bending stress of 165 MPa. For the combined test, with a 10 N force going into the page at the end of the handle, this yielded a bending stress of 166 MPa. The bending test yielded a stress of 18.6 MPa. Shear stress was calculated for all cases and yielded stresses under material yield strength.

Finite Element Analysis:

Figure 5 - Bending FEA

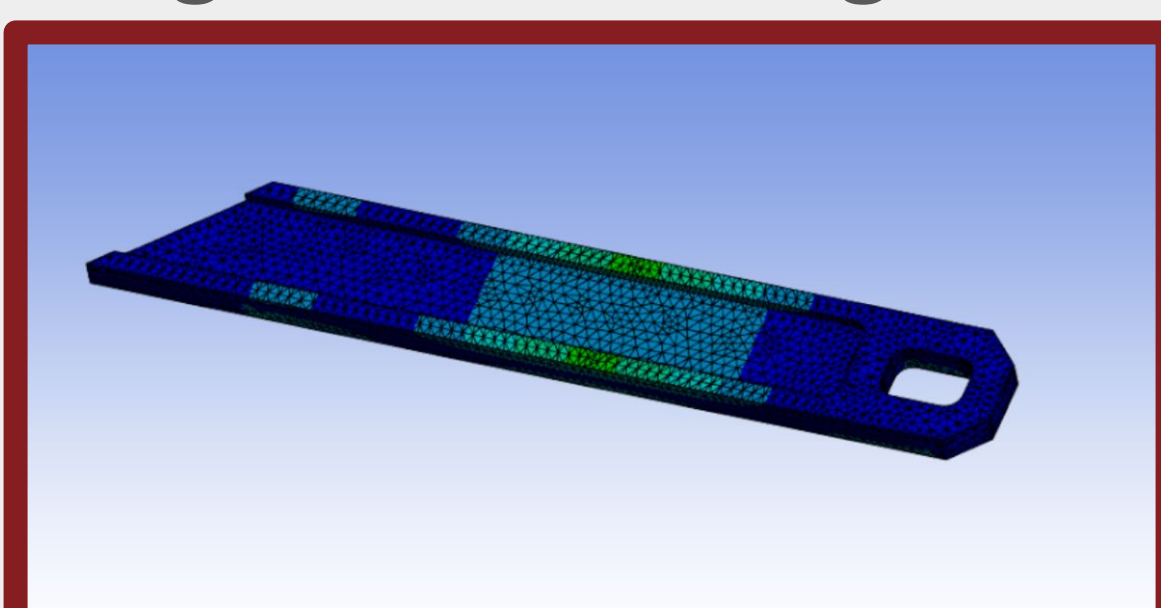
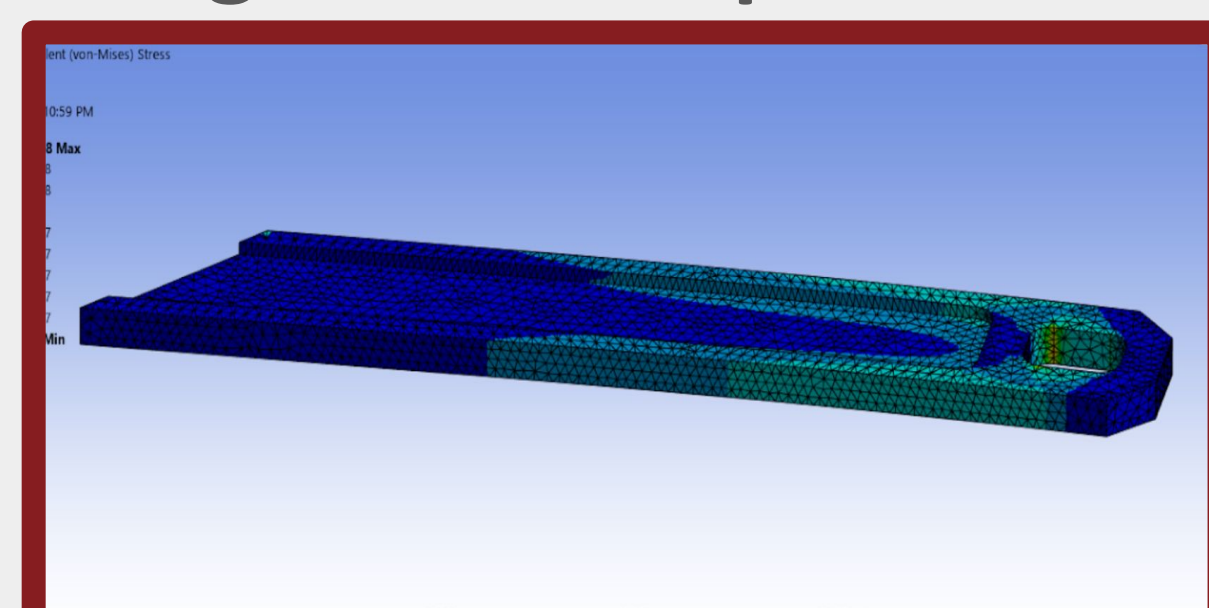


Figure 6 - Torque FEA



The final design performed exceptionally well during simulation. For the bending test, a max Von Mises equivalent stress of 26 Mpa was recorded. For the torque test, this max value was 45 Mpa. Finally, for the combined case, the max observation was 73 Mpa. Thus, the wrench operated with a safety factor greater than 3.5. Working towards this design, iterations succeeded in reducing weight and improving manufacturability.

Interpretation of Test Results:

The final weight of the wrench was 1.15 oz, a value greater than expected. This means that manufacturing operations left the product slightly oversized. The bending test was easily passed with non-permanent deformation. The torque test posed a significant challenge. Upon loosening the bolt, the wrench jaws stripped indicating plastic failure. This is in part explained by aluminum's soft material property in relation to the steel bolt. This is also explained by the clearance distance between the bolt and wrench points of contact. Too much clearance allowed for slippage causing increased localized forces at the bolt corners, which dug into the wrench upon applying torques. This circumstance hadn't been considered during the analysis stages.